R16

Code No: 134AK

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, November/December - 2020
COMPUTER ORGANIZATION
(Common to CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Draw the block diagram of micro program sequencer for a control memory and explain its operations in detail. [7+8]

2. Discuss the various Addressing modes in 8086. [15]

3. Briefly explain the Interrupt cycle of 8086. [15]

4. Explain the floating point addition- subtraction unit with a neat diagram. [15]

5.a) A non pipeline system takes 50 ns to process a task. The same task can be processed in a 6 segment pipeline with a clock cycle of 10 ns. Determine the speed up ratio of the pipeline for 100 tasks. What is the maximum speed up that can be achieved?
b) What are the application areas where vector processing is of utmost importance? Explain. [8+7]

6.a) With the help of neat diagram, explain the block diagram of digital computer.
b) Give a brief note on instruction codes. [8+7]

7. Explain the Machine language instruction formats in 8086. [15]

8. Explain the Interrupts and Interrupt service routines in 8086. [15]

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